

Reduction formulae

Basic skills

$$Q1. \int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

from $\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$

$$\begin{aligned} Q2. a) \int x \sin x dx & \quad \text{Let } u = x \quad v = -\cos x \\ & \quad \frac{du}{dx} = 1 \quad \frac{dv}{dx} = \sin x \\ & = -x \cos x - \int -\cos x dx \\ & = -x \cos x + \int \cos x dx \\ & = -x \cos x + \sin x (+c) \end{aligned}$$

$$\begin{aligned} b) \int x^5 \ln 2x dx & \quad u = \ln 2x \quad v = \frac{1}{6} x^6 \\ & \quad du = \frac{1}{x} \quad dv = x^5 \\ & = \frac{1}{6} x^6 \ln 2x - \int \frac{1}{6} x^5 dx \\ & = \frac{1}{6} x^6 \ln 2x - \frac{1}{36} x^6 (+c) \end{aligned}$$

Pick to diff
Logarithm

$$\begin{aligned} c) \int x \sqrt{x+2} dx & \quad u = x \quad v = \frac{2}{3} (x+2)^{\frac{3}{2}} \\ & \quad u' = 1 \quad v' = (x+2)^{\frac{1}{2}} \\ & = \frac{2}{3} x (x+2)^{\frac{3}{2}} - \int \frac{2}{3} (x+2)^{\frac{3}{2}} dx \\ & = \frac{2}{3} x (x+2)^{\frac{3}{2}} - \frac{4}{15} (x+2)^{\frac{5}{2}} (+c) \end{aligned}$$

$$d) \int 4x e^{\frac{1}{2}x} dx \quad \begin{array}{l} u = 4x \\ u' = 4 \end{array} \quad \begin{array}{l} v = 2e^{\frac{1}{2}x} \\ v' = e^{\frac{1}{2}x} \end{array}$$

$$= 8x e^{\frac{1}{2}x} - \int 8e^{\frac{1}{2}x} dx$$

$$= 8x e^{\frac{1}{2}x} - 16e^{\frac{1}{2}x} (+C)$$

$$Q3. a) \int x^2 \sin x dx \quad \begin{array}{l} u = x^2 \\ u' = 2x \end{array} \quad \begin{array}{l} v = -\cos x \\ v' = \sin x \end{array}$$

$$= -x^2 \cos x + \int 2x \cos x dx$$

$$b) \int 2x \cos x dx \quad \begin{array}{l} u = 2x \\ u' = 2 \end{array} \quad \begin{array}{l} v = \sin x \\ v' = \cos x \end{array}$$

$$= 2x \sin x - \int 2 \sin x dx$$

$$= 2x \sin x + 2 \cos x$$

$$c) \text{ so } \int x^2 \sin x dx = -x^2 \cos x + 2x \sin x + 2 \cos x (+C)$$

Competency skills

$$Q4. \quad I_n = \int x^n \sin x dx$$

$$a) \text{ using } \begin{array}{l} u = x^n \\ u' = nx^{n-1} \end{array} \quad \begin{array}{l} v = -\cos x \\ v' = \sin x \end{array}$$

$$\text{so } I_n = -x^n \cos x + n \int x^{n-1} \cos x dx$$

b) $\int x^{n-1} \cos x \, dx$ $u = x^{n-1}$ $v = \sin x$
 $u' = (n-1)x^{n-2}$ $v' = \cos x$

$$= x^{n-1} \sin x - (n-1) \int x^{n-2} \sin x \, dx$$

$$= x^{n-1} \sin x - (n-1) I_{n-2}$$

c) Hence

$$I_n = -x^n \cos x + n (x^{n-1} \sin x - (n-1) I_{n-2})$$

$$= -x^n \cos x + n x^{n-1} \sin x - n(n-1) I_{n-2}$$

Q 5. $I_n = \int x^n \cos x \, dx$ $u = x^n$ $v = \sin x$
 $u' = nx^{n-1}$ $v' = \cos x$

$$I_n = x^n \sin x - n \int x^{n-1} \sin x \, dx$$

$$\int x^{n-1} \sin x \, dx$$

$u = x^{n-1}$ $v = -\cos x$
 $u' = (n-1)x^{n-2}$ $v' = \sin x$

$$= -x^{n-1} \cos x + (n-1) \int x^{n-2} \cos x \, dx$$

$$= -x^{n-1} \cos x + (n-1) I_{n-2}$$

so $I_n = x^n \sin x - n (-x^{n-1} \cos x + (n-1) I_{n-2})$

$$\Rightarrow I_n = x^n \sin x + n x^{n-1} \cos x - n(n-1) I_{n-2}$$

Q6. $I_n = \int x^n e^x dx$ $u = x^n$ $v = e^x$
 $u' = nx^{n-1}$ $v' = e^x$

$$I_n = x^n e^x - n \int x^{n-1} e^x dx$$

$$I_n = x^n e^x - n I_{n-1}$$

Q7. $I_n = \int_0^k (k-x)^n \sin x dx$

a) $u = (k-x)^n$ $v = -\cos x$
 $u' = -n(k-x)^{n-1}$ $v' = \sin x$

so $I_n = \left[-(k-x)^n \cos x \right]_0^k - \int_0^k (k-x)^{n-1} \cos x dx$

$$I_n = (0 - -k^n) - \int_0^k (k-x)^{n-1} \cos x dx$$

$$\int_0^k (k-x)^{n-1} \cos x dx$$

$u = (k-x)^{n-1}$ $v = \sin x$
 $u' = -(n-1)(k-x)^{n-2}$ $v' = \cos x$

$$= \left[(k-x)^{n-1} \sin x \right]_0^k + (n-1) \int_0^k (k-x)^{n-2} \sin x dx$$

$$= (0 - 0) + (n-1) I_{n-2} = (n-1) I_{n-2}$$

so $I_n = k^n - n(n-1) I_{n-2}$

b) $\int_0^{\frac{3\pi}{2}} \left(\frac{3\pi}{2} - x \right)^2 \sin x dx$ i.e. I_2 with $k = \frac{3\pi}{2}$

so $I_2 = k^2 - 2 \times 1 I_0$

$$I_2 = \left(\frac{3\pi}{2}\right)^2 - 2I_0$$

$$I_0 = \int_0^{\frac{3\pi}{2}} \left(\frac{3\pi}{2} - x\right)^0 \sin x \, dx = \int_0^{\frac{3\pi}{2}} \sin x \, dx$$

$$= \left[-\cos x\right]_0^{\frac{3\pi}{2}} = (-0 - -1) = 1$$

so $I_2 = \frac{9\pi^2}{4} - 2 \times 1 = \frac{9\pi^2 - 8}{4}$

Advanced skills

Q8. $x^n \ln x$ doesn't need reduction as $\ln x$ differentiates to a power of x

$$\begin{aligned} \int \underbrace{x^n}_{v'} \underbrace{\ln x}_u \, dx &= \frac{1}{n+1} x^{n+1} \ln x - \int \frac{1}{n+1} x^n \, dx \\ &= \frac{1}{n+1} x^{n+1} \ln x - \frac{1}{(n+1)^2} x^{n+1} + c \end{aligned}$$

Howev $\ln^n x = (\ln x)^n$ doesn't go to a power of x

$$\begin{aligned} \int \underbrace{1 \times (\ln x)^n}_{v' \times u} \, dx &= x(\ln x)^n - \int \cancel{x} \frac{n x (\ln x)^{n-1}}{\cancel{x}} \, dx \\ &= x \ln^n x - n I_{n-1} \end{aligned}$$

Q9. $I_n = \int_0^4 \frac{x^n}{\sqrt{16-x^2}} dx$

NOTE.
improper as
undefined at
 $x=4$

a) $\int \frac{x}{\sqrt{16-x^2}} dx$ let $u = 16-x^2$
 $du = -2x dx$

$= \int -\frac{1}{2} \times \frac{1}{\sqrt{u}} du = -u^{\frac{1}{2}} = -\sqrt{16-x^2}$

b) I_n by part using $u = x^{n-1}$ $v = -\sqrt{16-x^2}$
 $u' = (n-1)x^{n-2}$ $v' = \frac{x}{\sqrt{16-x^2}}$

so $I_n = \left[x^{n-1} \sqrt{16-x^2} \right]_0^4 + (n-1) \int_0^4 x^{n-2} \sqrt{16-x^2} dx$
 $= (n-1) \int_0^4 x^{n-2} \sqrt{16-x^2} dx$

now $\int_0^4 x^{n-2} \sqrt{16-x^2} dx$

$= \int_0^4 \frac{x^{n-2}}{\sqrt{16-x^2}} (16-x^2) dx$

$= 16 \int_0^4 \frac{x^{n-2}}{\sqrt{16-x^2}} dx - \int_0^4 \frac{x^n}{\sqrt{16-x^2}} dx$

$$= 16 I_{n-2} - I_n$$

$$\text{so } I_n = (n-1)(16 I_{n-2} - I_n)$$

$$\Rightarrow n I_n = 16(n-1) I_{n-2}$$

$$\text{so } I_n = \frac{16(n-1)}{n} I_{n-2}$$

$$\begin{aligned} \text{c) } I_5 &= \frac{16 \times 4}{5} I_3 = \frac{16 \times 4}{5} \left(\frac{16 \times 2}{3} I_1 \right) \\ &= \frac{16^2 \times 8}{15} I_1 \end{aligned}$$

$$\begin{aligned} \text{now } I_1 &= \int_0^4 \frac{x}{\sqrt{16-x^2}} dx = \left[-\sqrt{16-x^2} \right]_0^4 \\ &= 0 - (-\sqrt{16}) = 4 \end{aligned}$$

$$\text{so } I_5 = \frac{16^2 \times 8}{15} \times 4 = \frac{8192}{15}$$

Q 10.

$$I_n = \int_0^{u^3} \tanh^n x \, dx = \int_0^{u^3} \tanh^2 x \tanh^{n-2} x \, dx$$

$$I_n = \int_0^{\ln 3} \tanh^{n-2} x (1 - \operatorname{sech}^2 x) dx$$

$$= \int_0^{\ln 3} \tanh^{n-2} x - \int_0^{\ln 3} \operatorname{sech}^2 x \tanh^{n-2} x dx$$

$$= I_{n-2} - \int_0^{\ln 3} \operatorname{sech}^2 x \tanh^{n-2} x dx$$

For $\int_0^{\ln 3} \operatorname{sech}^2 x \tanh^{n-2} x dx$ let $u = \tanh x$
 $u' = \operatorname{sech}^2 x$
 So $du = \operatorname{sech}^2 x dx$

$\Rightarrow = \int_0^{\frac{4}{5}} u^{n-2} du$ ← $\tanh(\ln 2) = \frac{4}{5}$

$$= \left[\frac{u^{n-1}}{n-1} \right]_0^{\frac{4}{5}} = \frac{1}{n-1} \left(\frac{4}{5} \right)^{n-1} - 0$$

$$\text{So } I_n = I_{n-2} - \frac{1}{n-1} \left(\frac{4}{5} \right)^{n-1}$$